6809 Forth — Open Items Checklist

Everything below was explicitly flagged during the build as incomplete, unverified, or deliberately deferred. Nothing here is a surprise — each item was named at the point it came up. This is a consolidated list to work from, not a new set of findings. Regenerated to reflect fixes and design decisions made since the original version.

Known bugs — resolved since the original checklist

- **`DUMP`'s partial-final-line bug** — fixed via `DUVALID` tracking (`25_tools_word_set.asm`). The ASCII column no longer reads past the valid bytes on a short final line.
- **`SUBSTITUTE`** — completed. It now performs a real bounds-checked copy (prefix / replacement / suffix) via a shared `SUBCOPY` helper, reusing `SEARCHW` for the match. Still scoped to a single registered name/value pair, not a full table — see "Open design questions" below; that scope limit is a deliberate simplification, not a bug.
- **`VALUE`'s PFA** — moved from `CODEHERE` (immutable space) to `VARHERE` (mutable space), matching `VARIABLE`'s pattern, so every `TO`-driven update now writes into the correct region instead of into code space. Not on the original checklist (found and fixed after this list was first generated); logged here for the record.
- **Every scratch/global cell now has a real `RMB`-assigned address in page zero** (`00_memory_map_and_globals.asm`), applied rather than left as a placeholder. Two real findings came out of doing this: the budget is **253 of 256 bytes used — only 3 bytes of headroom** in the GLOBALS page, and `SNEND` (documented in the original placeholder list) turned out to be genuinely dead — never read or written anywhere — and was dropped rather than given an address. Any future scratch cell added to this system will need to fit in that remaining 3 bytes or the page-zero fast-addressing property (DP = `$00`) stops covering it.
- **The ROM base dictionary is now real** (`27_forth_dictionary.asm`) — every primitive with actual code got a real header, chained via `LINK`, `CFA` pointing straight at its code label. `ABORTHDR`'s `LINK` field, a placeholder `0` since it was first built, is now resolved to the chain's newest entry. Building this surfaced two real findings, not just mechanical work: the original 1024-byte `BASEDICT` could not hold the header table (ultimately 1954 bytes needed, once `DOES>` was added — see below), so it was resized to 2048 bytes, taking the space from `BASECODE` (now 6080 bytes, was 7104) — a resize based purely

on the header-table budget, not on any actual measurement of `BASECODE`'s assembled size, which has never been checked with a real 6809 assembler. Generating this table also surfaced that **`DOES>` had no corresponding code anywhere in the file** — resolved in a follow-up pass, see below.

- **`DOES>` — resolved.** `SETDOES` (the patching runtime) was added beside `DODOES` / `DOESRT0`, and `DOES>` itself (code label `DOESGT`, since a literal `">"` is not a valid 6809 assembler label) was added right after `CREATE`. `DOESBEH` was added to the GLOBALS layout to support it, using 2 of the page's last 3 free bytes — only 1 byte of headroom now remains in page zero. `DOES>` also has a real dictionary header now (`H_DOESGT`, the chain's newest entry); `ABORTHDR`'s `LINK` was updated again to point at it.
- **Four internal loop-label names were reused across unrelated routines: `FLOOP`, `UDLOOP`, `DRPOS`, `CMLOOP`.** Resolved — each pair renamed to a distinct name scoped to its own routine: `FIND`'s `FLOOP` → `FFLOOP`; `FILLW`'s `FLOOP` → `FILLOOP`; `UDIV16`'s `UDLOOP` → `UD16LOOP`; `UDDIGIT`'s `UDLOOP` → `UDDLOOP`; `DIVCOMMON`'s `DRPOS` → `DCRPOS`; `DDOTR`'s `DRPOS` → `DDRPOS`; `CMOVEW`'s `CMLOOP` → `CMVLOOP`; `COMPAREW`'s `CMLOOP` → `CMPLOOP`. Verified zero duplicate labels and zero stray references to any of the old names remain anywhere in the file.
- **Hardware (RTS) serial handshaking — implemented.** `IRQH`'s receive path and `KEY` now toggle RTS based on the input ring's fill level (`INFILL`, `RTSCHECKHI`, `RTSCHECKLO` against `INHIWATER`=48/`INLOWATER`=16, with hysteresis between them) via a new `CR_RTSHI` control byte and `RTSSTATE` flag — the last free byte in the GLOBALS page, which is now fully packed at 256/256. CTS (the other handshaking direction) needed no firmware logic at all: confirmed against the 6850 datasheet that TDRE is automatically inhibited while CTS is deasserted, so this system's existing TDRE-gated transmit logic already respects it. A real race was identified and closed: mainline code (`KEY`, via `RTSCHECKLO`) and the ISR (`IRQH`'s `TXOFF`) can both decide to write the control register, so `RTSCHECKLO` masks IRQ around its critical section and `TXOFF` checks `RTSSTATE` before writing. Software (XON/XOFF) handshaking remains unimplemented — see below.
- **`TRUE` and `FALSE` — implemented.** Surfaced while organizing the ANS test suite: neither existed as an actual dictionary word, only as the internal `TRUEV` / `FALSEV` assembler constants. Added as `CONSTANT TRUE -1` / `CONSTANT FALSE 0` (`TRUEBODY` / `FALSEBODY`, section 26; headers `H_TRUE` / `H_FALSE`, chain's two newest entries, section 27) — the first `CONSTANT`-pattern ROM-resident words in this system. Every other ROM word's CFA is a plain code label; a `CONSTANT`'s CFA is the `DODOES`-trampoline pattern instead, built here with fixed, assemble-time addresses rather than a `CODEHERE` snapshot, since there is no interactive `CREATE` / `CONSTANT` phase for ROM content.

Structural duplication (identified, some resolved, some not)

- `:` / `CREATE` / `VARIABLE` 's header-building — resolved via `HEADER` (section 7).
- `COMMA` / `CODECOMMA` / `CCOMMA` / `VCOMMA` / `VCCOMMA` / `CCOMMA1` — resolved via `APPENDCELL` / `APPENDBYTE` (section 6).
- `<#` / `#>` recomputing PAD's address inline — resolved, both now call `PADW` (section 20 / section 6).
- No further known duplication, but the codebase was never given a full pass specifically hunting for more.

Real, load-bearing gaps

- **Software (XON/XOFF) handshaking is still not implemented.** Now that hardware RTS/CTS handshaking is in place, this is the one remaining flow-control gap: this system still never transmits XON/XOFF and would not recognize either byte as a control signal if the remote device sent them — an incoming `$11` / `$13` is just queued as an ordinary character. Only relevant for links with no RTS/CTS wiring (plain 3-wire serial); would need a byte- interception layer between the input ring and `KEY` 's caller.
- **No hard boundary checks** anywhere `DPHERE` / `CODEHERE` / `VARHERE` grow toward each other or toward the stacks. `UNUSED` / `VUNUSED` *report* the distance to each boundary but nothing enforces it — a runaway compile can silently corrupt an adjacent region.
- **`ENVTABLE` is incomplete.** Missing entries: `/HOLD` , `/PAD` (this build never fixed a capacity for either — filling them in would mean deciding a real bound first, not just picking a number), `MAX-D` , `MAX-UD` (need the dispatcher extended to push *two* cells for double-cell answers — current `ENVQUERY` only handles one), `WORDLISTS` / `FLOORED` (need the dispatcher to be able to report a recognized-but-false answer, distinct from "unrecognized name" — no such path exists yet).
- **`CATCH` -wrapped `QUIT` / `INTERPRET` rollback is scoped to one input line only.** A colon definition spanning multiple lines that fails partway through a later line only rolls back that line's contribution, not the whole definition back to `:` . A complete fix needs the region-pointer snapshot taken once at `:` and held until `;` or an error, not refreshed every `QLOOP` pass.
- **`ABORTHDR` 's `LINK` field is a placeholder `0`** in the consolidated/split source — must be set to the real prior `LATEST` value once final ROM layout/assembly order is fixed.

Never resolved after being explicitly raised

- **`J` doesn't generalize past one level of loop nesting.** No `J2`/deeper equivalent exists. A triple-nested loop's innermost body has no built-in way to reach the outermost index without manually replicating `J`'s offset arithmetic one frame further.
- **`LEAVE`'s correctness depends on always being textually inside the loop it affects** — never verified against a `LEAVE` called from a separately-defined word invoked from within a loop (which would read/set the wrong stack frame, or crash).
- **`WORD`'s 31-character cap is now inconsistent within the system.** `PARSE`/`PARSE-NAME` were deliberately rewritten to not share it, but `WORD` itself — and everything still built on it (`CHAR`, `[CHAR]`, header names, `FIND`) — still has it.
- **The optional Search-Order word set is not implemented.** Every word resolves through one unified dictionary chain rooted at `LATEST`; there is no multi-wordlist support (`WORDLIST`, `GET-ORDER`/`SET-ORDER`, `ALSO`/`ONLY`, etc.). This was a foundational decision fixed since `COLDSTRT`'s first version, not an oversight — now documented explicitly (ClaudeForth documentation, Section 4.5) along with what adding it later would actually require: restructuring `FIND` into a loop over an active search order, changing `HEADER` to link into a selectable "current" wordlist instead of unconditionally into `LATEST`, and fixing the few words (`RECURSE`, `;`'s unsmudge step, `WORDS`) that read `LATEST` directly today.

Open design questions (not defects — deliberate unresolved choices)

- Whether `ALLOT`/`VALLOT`/`PICK`/`ROLL`/`BASE` should range-check their inputs against actual stack/region bounds. Currently none of them do, consistently, by choice rather than oversight.
- Whether any additional distinction like `TIB`/`SOURCE` (fixed terminal buffer vs. current input source) is needed anywhere else in the system where `EVALUATE`'s redirection might still cause a mismatch.
- `SWIH`'s throw code (`-99` in the consolidated source) was never assigned a real, deliberate value — it's a placeholder for "some hardware trap occurred," not a considered ANS-style code.
- `REPLACES`/`SUBSTITUTE` remain single-slot (one registered name/value pair, overwritten by the next `REPLACES` call) rather than a true multi-entry table. A real table needs its own storage layout and lookup structure — a deliberate scope decision made when

`SUBSTITUTE` was completed, not an oversight.

What's solid (for reference, not action)

Stack manipulation (Core + Core Ext + return-stack transfer), all arithmetic (single + double + mixed precision), logic, comparison (single + double), full control flow including `CASE`, all defining words including `DEFER` / `MARKER` / `VALUE` / `TO` (`VALUE` now correctly targeting mutable space), memory and string operations (including a completed `SUBSTITUTE`), `SOURCE` / `EVALUATE` / `REFILL` mechanics, and the Tools word set (`.S` / `WORDS` / `DUMP` , with `DUMP` 's edge-case bug fixed) are complete and were traced/verified against concrete cases during the build, not merely asserted correct.